Evaluating 30-Day Hospital Readmission Prediction in Patients with Diabetes: Patient-Level Versus Encounter-Level Validation

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***Abstract*—Performance estimates for readmission prediction depend on whether repeated encounters from the same patient can cross evaluation partitions. We investigate this dependence using the Diabetes 130-US Hospitals dataset. A reproducible audit identifies 99,343 eligible encounters from 69,990 patients after excluding death and hospice discharges; 11.39% have the positive readmission label. We compare L2 and L1 logistic regression, random forest, and LightGBM under patient-disjoint and stratified encounter-level evaluation. Each protocol is repeated across five predefined seeds, with five-fold development-set validation and training-only preprocessing. Average precision is the primary outcome. The four weighted models achieve mean patient-level AP between 0.208 and 0.228; LightGBM has the highest mean. Encounter-level test sets contain 41.7% encounters from patients represented in development data on average. These initial experiments quantify evaluation sensitivity without treating a difference between distinct test populations as causal evidence of leakage. Calibration correction, resampling, and detailed subgroup analyses remain planned. This cumulative manuscript reports completed implementation, responds to prior feedback, and defines the remaining analysis.**

***Index Terms*—readmission, diabetes, grouped validation, repeated encounters, average precision**

I. Introduction

Hospital readmission prediction is a practical application of data mining to structured clinical records. At discharge, information about diagnoses, treatment, prior utilization, and the completed hospital stay could help identify encounters associated with a subsequent admission. Nevertheless, performance on a historical benchmark does not establish that a model will improve care. A useful initial objective is to determine what the available data and evaluation design actually support.

The Diabetes 130-US Hospitals dataset contains inpatient encounters collected during 1999–2008 [1], [2]. An encounter is not an independent patient: individuals may contribute several records. An encounter-level random split can therefore evaluate a mixture of previously represented and unseen patients, whereas a patient-level split evaluates individuals absent from development data. Removing patient identifiers from predictors does not remove this dependence, because clinical and utilization characteristics may recur across visits. Conversely, the presence of overlap alone does not quantify its effect on performance.

Earlier work has examined clinical associations [2], compared predictive algorithms [3], and explicitly incorporated patient-level partitioning and training-only resampling [4]. The practical need addressed here is a reproducible, repeated comparison of validation protocols for a fixed cohort and common model families. We do not claim a new algorithm or that grouped splitting itself is novel. Nor do we assume that encounter splitting must produce higher scores.

Our questions are: **RQ1**, how well do four established models rank positive encounters under patient-disjoint evaluation? **RQ2**, how do average precision and model rankings change under encounter-level evaluation across seeds? **RQ3**, how much patient overlap occurs, and how do probability quality and performance within seen/unseen patient subsets qualify the comparison? Prediction is defined at discharge, not at admission; the main target is generalization to unseen patients within this historical dataset.

The specific contributions of this report are (1) a verified cohort and repeated-patient audit, including separate treatment of unperformed laboratory tests; (2) a reproducible comparison of four model families under two validation protocols and five seeds; and (3) an uncertainty-aware interpretation of overlap, ranking, and calibration. The proposed audience remains the IEEE Intermountain Engineering, Technology, and Computing Conference identified in Report 1; IEEE conference format is retained.

II. Related Work

A. Clinical analysis and predictive benchmarking

Strack et al. [2] restricted their analysis to the first encounter per patient and excluded death/hospice outcomes. This avoids repeated observations by changing the cohort and supports their analysis of HbA1c testing and readmission. Our study retains repeated encounters to investigate evaluation dependence explicitly. The two designs address different questions, so their sample size and reported associations are not interchangeable with our predictive results.

Emi-Johnson and Nkrumah [3] compare logistic regression, random forest, XGBoost, and a neural network using a stratified 80/20 split, class weighting, cross-validation, and SHAP. Patient-disjoint partitioning is not explicitly established in the reported splitting procedure. That is a reporting distinction, not proof that their implementation leaked information. Their work

motivates model comparison, while our emphasis is partition sensitivity rather than cross-paper score superiority.

B. Leakage safeguards and the closest comparison

Afridi et al. [4] explicitly describe patient-level partitioning, fold-specific preprocessing and feature selection, SMOTENC/class weighting, and SHAP/LIME explanations. Their analytical cohort contains 75,081 encounters, and their partition/training description specifies seed 42. These safeguards are prior work. Our addition is a repeated encounter-versus-patient comparison, measured overlap, and model-ranking stability on a consistently defined cohort. We do not attempt a numerical replication of their different cohort or neural architectures. Table I summarizes the methodological distinctions; "not explicit" means not established by the reported methods.

C. Imbalance and evaluation

Precision–recall analysis is useful when positives are uncommon [5]. We operationalize it as non-interpolated average precision (AP), rather than trapezoidal PR area [6]. A constant-score reference makes test prevalence visible. Group-aware validation addresses dependence between observations from the same person [7]. For imbalance, weighting and resampling are distinct choices: both can change the meaning of fitted probabilities, so ranking performance alone is insufficient. We include an unweighted logistic reference and Brier scores; SMOTENC through imbalanced-learn [8] remains a planned comparison. LightGBM [9] supplies a computationally feasible nonlinear model alongside random forest and regularized logistic regression.

III. Data and Preprocessing

A. Source, cohort, and outcome

We downloaded the public UCI files [1], including their categorical-ID mapping, and recorded a SHA-256 checksum. The CSV has 101,766 rows and 50 columns: two identifiers, the outcome, and 47 candidate predictors. The unit of analysis is one inpatient encounter. The outcome is $y = 1$ for `<30`, and $y = 0$ for `>30` or `NO`; no additional readmission labels were inferred.

Fixed eligibility rules exclude discharge codes 11, 13, 14, 19, 20, and 21 (death or hospice categories in the supplied mapping). They remove 2,423 rows, leaving 99,343 encounters, 69,990 patients, and 11,314 positives (11.39%). Code 21 has no observed rows but is retained in the rule. All age categories and unknown/invalid gender records are retained; no unreported adult-only filter is applied. Transfers remain eligible in this version, with their discharge category represented as a predictor. This decision requires a later sensitivity analysis because transfers need not represent an ordinary community discharge.

B. Completed quality audit

The original data contain zero exact duplicate rows, zero repeated encounter IDs, and zero duplicate rows after removing the two identifiers. Repeated patients are a separate phenomenon: 16,341 patients (23.35% of eligible patients) contribute 45,694 encounters (46.00% of eligible encounters). The median is one encounter, the 90th percentile two, the 95th three, the 99th six, and the maximum 40. Figure 1 shows the distribution. These observations support preserving patient groups rather than deleting repeat visits.

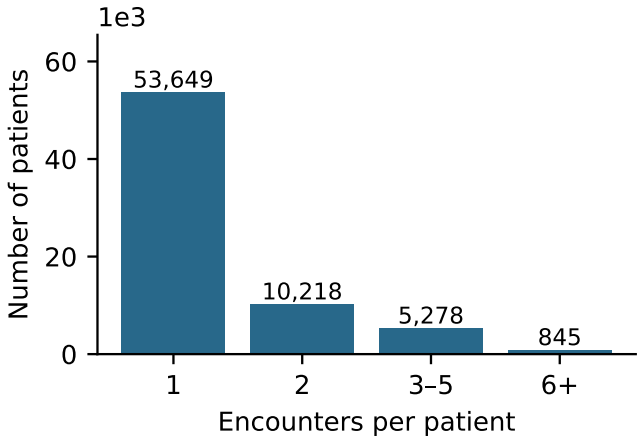


Fig. 1. Eligible patients grouped by encounter count. Repeat encounters are retained and grouped during primary evaluation.

Table II separates structural test absence from unknown values. In particular, `None` in HbA1c/glucose fields is read literally and recoded as "Not performed," consistent with the dataset definitions [2]; it is not numerically imputed. The `?` marker becomes an explicit "Unknown" category. Weight is excluded according to the pre-existing plan. Unknown diagnosis counts are 20, 356, and 1,419 for the three diagnosis fields; they remain explicit unknown categories. Numerical ranges and category counts are exported for review. No outcome-dependent outlier deletion or winsorization is performed.

C. Representation and safeguards

The model input has 46 variables after removing weight and both IDs. Eight count/duration variables are standardized using training-fold means and standard deviations. The remaining variables, including age bands and admission/discharge IDs, are categorical. ICD-9 diagnoses are mapped by fixed rules into nine clinical groups plus unknown, adapting the grouping in [2] to all three diagnosis fields. The mapping treats supplemental E/V codes as other and preserves unknown codes separately. Categorical vocabulary is learned through training-only one-hot encoding; unseen categories are ignored at transformation time. No numerical missing values require imputation in this cohort.

Patient and encounter IDs are used for grouping, assertions, and saved split manifests only. Fixed eligibility rules and semantic mappings precede splitting; all learned operations follow it (Fig. 2). No longitudinal features are reconstructed from future encounters. The supplied prior-utilization variables are used as documented. Because usable encounter dates and hospital group identifiers are unavailable, patient-disjoint

TABLE I
COMPARISON OF RELEVANT STUDIES AND THIS REPORT. MISSING REPORTING IS NOT EVIDENCE OF AN INCORRECT IMPLEMENTATION.

Study	Repeated patients	Validation	Leakage/imbalance handling	Relation to our study
Strack et al. [2]	First encounter per patient retained	Association modeling; not the same predictive holdout comparison	Death/hospice exclusions; clinical covariate analysis	Dataset semantics and cohort rationale
Emi-Johnson and Nkrumah [3]	Patient grouping not explicit in split description	Stratified 80/20; five-fold model tuning	Preprocessing pipeline and class weighting reported	Conventional predictive benchmark
Afridi et al. [4]	Patient groups kept separate	Patient holdout; five-fold grouped validation; seed 42 specified	Training-fold preprocessing, selection and SMOTENC/weights	Closest framework; safeguards already established
This report	All eligible encounters retained	Two protocols, five seeds; matched search budgets	Training-only fitting; overlap assertions; no test resampling	Quantifies protocol sensitivity and ranking variability

TABLE II
COMPLETED AUDIT OF THE ELIGIBLE COHORT. UNKNOWN VALUES ARE DISTINCT FROM TESTS NOT PERFORMED.

Item	Count (percentage)
Eligible encounters / patients	99,343 / 69,990
Positive / negative outcomes	11,314 / 88,029
Patients with repeat encounters	16,341 (23.35%)
Encounters from repeat patients	45,694 (46.00%)
Unknown weight	96,218 (96.85%)
Unknown medical specialty	48,616 (48.94%)
Unknown payer code	39,398 (39.66%)
Unknown race	2,234 (2.25%)
HbA1c not performed	82,509 (83.05%)
Glucose serum test not performed	94,191 (94.81%)

validation does not establish temporal or new-hospital generalization.

IV. METHODOLOGY AND IMPLEMENTATION

A. Models and objectives

A constant predictor outputs the development-set prevalence. L2 logistic regression is the main model baseline; L1 logistic regression evaluates sparse coefficient selection through an LASSO-type penalty [10]. For both, class weights are $w_k = n/(2n_k)$ using only the current training labels. Logistic models minimize a regularized weighted binary log loss, with inverse penalty strength C selected from $\{0.1, 1\}$ for L2 and $\{0.01, 0.1\}$ for L1. L1 and L2 penalties differ in whether coefficients are encouraged toward exact zero or smoothly shrunk. We use the liblinear solver and a 1,000-iteration limit. An otherwise equivalent unweighted L2 model provides a probability-quality reference.

Random forest [11] aggregates 100 trees with maximum depth 12 and square-root feature subsampling; minimum leaf size is selected from $\{5, 20\}$. LightGBM uses 150 boosting rounds, learning rate 0.05, minimum leaf count 50, and leaves selected from $\{15, 31\}$. Both use training-derived balanced class weights. These deliberately small, prespecified searches control runtime and provide equal two-candidate budgets rather than an exhaustive benchmark. No early stopping uses test data. Model configurations and selected values are exported per run.

B. Completed versus planned components

The data audit, semantic preprocessing, sparse encoding/scaling, four weighted models, unweighted logistic reference, two splitting protocols, repeated tuning/evaluation, and output generation are implemented. L1 performs embedded feature selection; no separate RFE step has been run. SMOTENC, post-hoc probability recalibration, SHAP/permutation interpretation, and discharge/subgroup ablations remain planned. Thus their benefits are not asserted in this report.

The environment uses Python, NumPy, pandas, SciPy, scikit-learn, LightGBM, and Matplotlib [12], [13], [14], [15], [9], [16]. Exact installed versions, platform information, and the data checksum are stored with the results. Sparse encoding reduces memory use; random forest receives a dense float32 matrix for faster fitting. Model libraries are limited to two threads. A timing pilot motivated a narrower L1 search before test scoring; no test result guided this change. No GPU, external model API, or private clinical data is required. The repository supplies a dataset downloader, experiment runner, summaries, split manifests, predictions, and manuscript source:
https://github.com/Ahmadjajja/Hospital_Readmission_Prediction.

V. EXPERIMENTAL DESIGN

A. Repeated outer splits and internal selection

For each seed in $\{11, 23, 42, 67, 101\}$, GroupShuffleSplit holds out 20% of unique patients. All encounters of those patients enter the test set, so the fraction of encounters is only approximately 20%. The comparison uses stratified random splitting of 20% of encounters. Each arm independently creates five development folds: StratifiedGroupKFold for the patient arm and StratifiedKFold for the encounter arm. Assertions verify zero patient intersection in every primary outer and inner split.

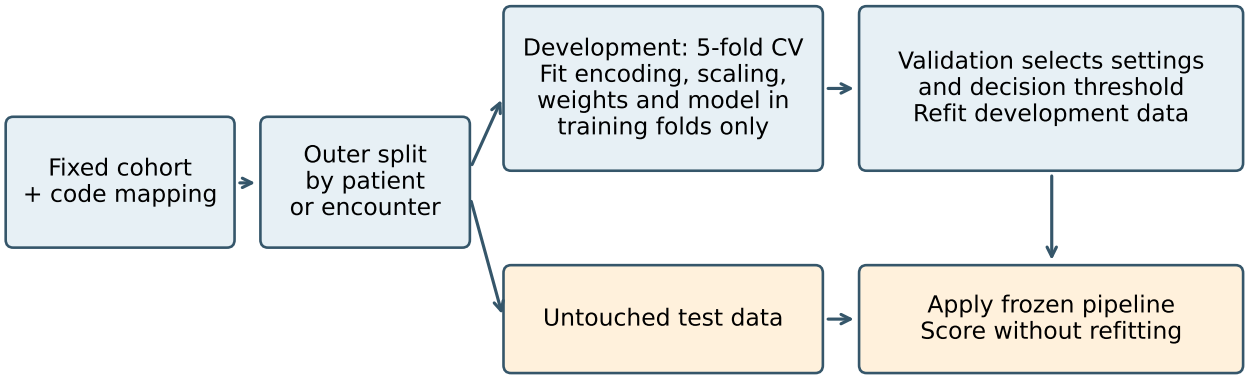


Fig. 2. Implemented workflow. The test branch bypasses all fitting and selection. L1 selection occurs inside training folds. SMOTENC is planned, not used in the reported runs; when added it will be confined to training folds before final encoding.

Each candidate is evaluated by mean fold AP. The selected candidate is refitted on the full development set. The F1-maximizing threshold is chosen from its pooled out-of-fold development predictions, then frozen for the test evaluation. These development scores are used for selection, not presented as unbiased generalization estimates. The same candidate sets and preprocessing rules are used across arms, but independently selected parameters may differ. This estimates the effect of the complete validation protocol rather than a pure causal effect of outer-split overlap. Test results do not feed back into tuning.

B. Metrics and uncertainty

Average precision is computed as

$$\mathrm{AP} = \sum_j (R_j - R_{j-1}) P_j, \tag{1}$$

where P_j and R_j are precision and recall along the score-ranked curve [6]. We also report ROC-AUC, precision, recall, F1, and Brier score $n^{-1}\sum_i (p_i - y_i)^2$. Threshold tuning affects the threshold-dependent measures, not AP or ROC-AUC. Accuracy is not a success criterion because an always-negative classifier exploits the imbalance.

Across seeds, we report means and sample standard deviations. The repetitions reuse patients and are not independent population samples; their standard deviation describes partition/model variability, not a confidence interval for clinical deployment. For a prespecified illustrative patient holdout (seed 42), 1,000 patient-cluster bootstrap samples (bootstrap RNG seed 2026) estimate percentile 95% intervals for AP differences against L2. All encounters of a sampled patient receive the same multiplicity, and all models use identical sampled clusters. These intervals are conditional on fitted models and that holdout; they do not incorporate all training uncertainty or adjust for multiple comparisons. We do not apply paired intervals across distinct protocol test sets.

C. Overlap and success criteria

For each encounter-split test set, we measure the fraction of test encounters whose patient appears in development data and count overlapping patients. We separately summarize seen/unseen subsets, including prevalence and sample size. These subsets differ in case mix and repeat frequency, so their differences are descriptive. A credible outcome is a complete, reproducible comparison with uncertainty, even when a complex model does not improve on the baseline or the split gap is small. Evidence of predictive improvement requires both effect size and uncertainty, not a predetermined score target.

VI. Implementation Progress and Initial Findings

A. Completed experiments and performance

All ten outer partitions completed: 500 candidate fits, 50 selected model fits, and ten constant-reference evaluations. The exported metrics contain 60 evaluated model/protocol/seed combinations. Automated checks recompute metrics from predictions, verify labels against source records, and verify partition integrity. No convergence warnings were observed in the completed-run logs. Table III and Fig. 3 summarize the main results. Values are preliminary benchmark findings, not clinical validation.

LightGBM has the highest mean patient-level AP (0.228), compared with 0.216 for weighted L2 and 0.112 for the constant reference. The encounter-minus-patient differences in mean AP are +0.0055 (L2 logistic), +0.0062 (L1 logistic), +0.0132 (Random forest), +0.0102 (LightGBM). These are descriptive differences across separately constructed test sets. They are not paired causal effects of overlap.

B. Overlap and conditional uncertainty

Patient-level holdouts have zero patient overlap by assertion. Encounter-level overlap ranges from 41.34% to 42.10% of test encounters (mean 41.70%). Patient-arm test prevalence ranges from 10.99% to 11.41%; the corresponding test sizes range

TABLE III
TEST PERFORMANCE, MEAN ± SAMPLE SD ACROSS FIVE SEEDS. CONSTANT AP EQUALS EACH TEST SET'S POSITIVE FRACTION.

Model	Split	AP	ROC-AUC
L2 logistic	Patient	0.216 ± 0.006	0.663 ± 0.003
L2 logistic	Encounter	0.221 ± 0.002	0.669 ± 0.005
L1 logistic	Patient	0.215 ± 0.006	0.665 ± 0.003
L1 logistic	Encounter	0.222 ± 0.002	0.671 ± 0.004
Random forest	Patient	0.208 ± 0.002	0.665 ± 0.005
Random forest	Encounter	0.221 ± 0.002	0.673 ± 0.004
LightGBM	Patient	0.228 ± 0.006	0.673 ± 0.002
LightGBM	Encounter	0.238 ± 0.004	0.679 ± 0.003
L2 unweighted	Patient	0.216 ± 0.006	0.662 ± 0.003
L2 unweighted	Encounter	0.221 ± 0.002	0.668 ± 0.005
Constant	Patient	0.112 ± 0.002	0.500 ± 0.000
Constant	Encounter	0.114 ± 0.000	0.500 ± 0.000

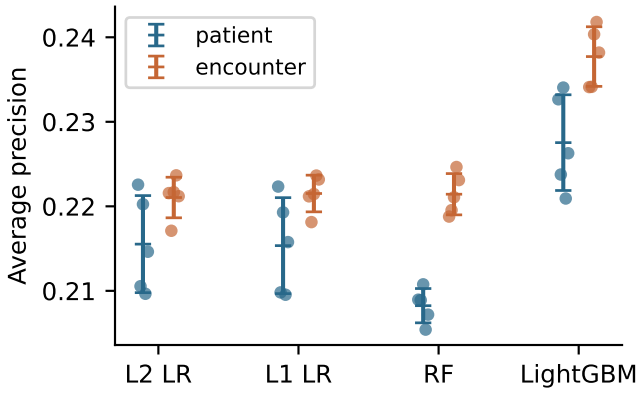


Fig. 3. AP across five seeds. Points are individual runs; marks and error bars show means and sample standard deviations.

from 19,717 to 19,991. The constant-reference results make this variation explicit.

For LightGBM within encounter-level holdouts, mean AP is 0.279 for seen patients and 0.157 for unseen patients. Their mean positive fractions are 20.10% and 5.16%, respectively. The case-mix and prevalence difference prevents interpreting this subset contrast as an isolated overlap effect.

For the prespecified patient holdout at seed 42, paired AP differences relative to weighted L2 and their 95% patient-bootstrap intervals are L1 logistic: -0.0010 [-0.0026, +0.0006]; Random forest: -0.0095 [-0.0169, -0.0021]; LightGBM: +0.0138 [+0.0073, +0.0214]. These conditional, unadjusted intervals complement rather than replace the repeated-seed summaries.

C. Thresholds and probability quality

Table IV reports the precision–recall trade-off after development-only threshold selection. Unweighted L2 has mean Brier score 0.095, compared with 0.225 for weighted L2. The calibration diagnostic (Fig. 4) illustrates why class-weighted scores should not automatically be interpreted as calibrated risks. Brier score combines discrimination and calibration; a lower value alone does not establish perfect calibration. No probability correction has yet been fitted.

TABLE IV
PATIENT-LEVEL THRESHOLD AND PROBABILITY DIAGNOSTICS: MEANS ACROSS FIVE RUNS. THRESHOLDS MAXIMIZE DEVELOPMENT OUT-OF-FOLD F1.

Model	Precision	Recall	F1	Brier
L2 logistic	0.193	0.475	0.274	0.225
L1 logistic	0.193	0.476	0.275	0.225
Random forest	0.189	0.500	0.274	0.219
LightGBM	0.211	0.435	0.284	0.215
L2 unweighted	0.197	0.447	0.272	0.095
Constant	0.000	0.000	0.000	0.099

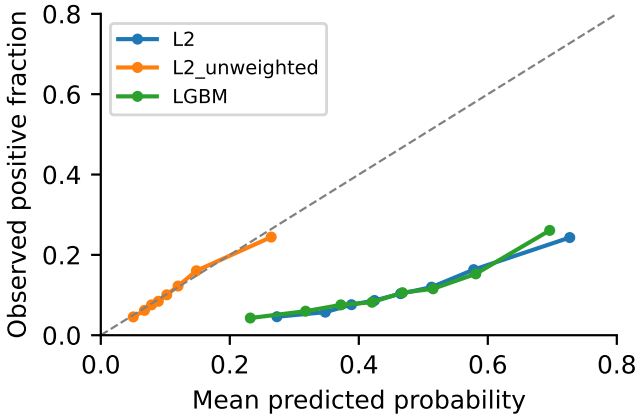


Fig. 4. Calibration diagnostic for the prespecified patient holdout, seed 42. Eight quantile bins; dashed line indicates ideal calibration. This is a diagnostic, not a fitted correction.

VII. DISCUSSION, LIMITATIONS, AND RESPONSIBLE USE

The central distinction is between generalizing to unseen patients and scoring a mixture of seen and unseen patients. An encounter-level score is not automatically invalid for every application, but it does not directly answer the unseen-patient question. Neither protocol simulates prospective deployment because the files lack usable chronology. The diagnostic comparison can reveal sensitivity; it cannot establish that overlap alone caused a score difference.

Cohort definitions differ from prior studies, including removal of first-encounter restrictions and retention of transfer discharges. Cross-paper performance rankings would therefore be misleading. The limited search spaces may favor some models, and repeated encounters give frequent users greater weight in encounter-level metrics. A patient-weighted sensitivity analysis is planned. Out-of-system readmissions may be unobserved, and historical practice, coding, and treatment patterns may not represent current care.

UCI distributes the dataset under CC BY 4.0 [1]. The data are deidentified, but demographic fields remain sensitive; we make no reidentification attempt. The repository documents the source and license. Race, gender, and age categories may reflect uneven representation and measurement; subgroup calibration and error analysis remain unfinished. No fairness, causal, intervention-benefit, or clinical-readiness claim is

made. Feature importance, when added, will describe model associations rather than treatment effects.

VIII. Response to Feedback and Updated Outline

Figure 2 replaces the earlier ambiguous workflow with a full-width figure and a separate untouched-test branch. Learned preprocessing and L1 selection appear after splitting; planned resampling is explicitly training-only. Five predefined seeds and variability summaries address partition stability. Table I supplies the requested literature comparison and incorporates Afridi et al., distinguishing our protocol-comparison objective from their framework. The duplicate and encounter-frequency audit is now completed, and Table II distinguishes unperformed tests from unknown values. Completed and planned components are labeled, and references use a consistent numbered IEEE style. The research question and dataset remain unchanged; an unweighted logistic reference was added to clarify calibration effects.

The final paper outline remains: abstract; introduction; related work; data/preprocessing; methodology; experimental design; results; discussion; limitations/responsible use; conclusion; references. Sections I–VI now contain complete prose and initial results. The final Results section will add the imbalance ablation, probability calibration comparison, subgroup tables, and feature-importance figure. Discussion will integrate those outputs and assess consistency across seeds. The final Conclusion will report the supported scope of generalization and unresolved limitations. No favorable result is assumed for these pending sections.

IX. Remaining Work and Responsibilities

The next milestone is Progress Report 3; its exact due date was not supplied. The operational schedule below uses target windows rather than claiming an instructor-approved deadline. Ahmad coordinates implementation and reproducibility; Muhammad Shehzore Hashmi coordinates validation and statistical checks; Akshay Kumar Dosapati coordinates literature synthesis and figures. These are planned coordination responsibilities, not a claim that any member independently completed a particular run. All members will review the code, source papers, model outputs, and manuscript.

If resampling exceeds resources, restrict it to one model and report that scope. If subgroup positives are insufficient, report counts without unstable comparative claims. If no model improves materially, retain the negative finding. Any protocol changes after this report will be documented and distinguished from the current prespecified runs; reuse of these test sets for later exploratory analyses will not be represented as a new untouched confirmatory evaluation.

TABLE V
Remaining work toward Report 3; dates are internal targets.

Target	Task and dependency	Coordinator
Nov. 9–10	Verify all runs and cohort rules; inspect transfer/unknown discharge categories	Shehzore, all
Nov. 11–12	Training-only SMOTENC comparison on a limited model set; requires mixed-type pipeline	Ahmad
Nov. 13–14	Calibration correction from development data; discharge and patient-weighted sensitivity	Ahmad, Shehzore
Nov. 15–16	Subgroup counts/errors and feature-importance figure; sufficient positives required	Akshay, all
Before R3	Integrate literature, ablations, interpretation, and final reproducibility review	All members

X. Preliminary Conclusion

The project has progressed from a blueprint to a verified audit and executable repeated evaluation. Repeated encounters are substantial, while duplicate records are absent under the checks performed. The implemented comparison now measures model performance and protocol variability with consistent safeguards. The final study will extend these initial findings with focused imbalance, calibration, and interpretation analyses rather than adding unsupported claims of clinical utility.

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